Supplementary Information

Mapping the AI Research of China, the US and the EU:

A Scientometric Analysis of Citation Shares within AI Subdisciplines (2020–2024)

Julius Pfundstein Thomas Efer Manuel Burghardt

Contents

This document holds the complete cluster-level results that the manuscript summarises. It has two parts:

- **Section 1** — the fractional citation share held by each bloc within every cluster of the taxonomy, at all three hierarchy levels. The manuscript shows these shares graphically (Figure 2) and lists the domains and fields in its Appendix Table A1; these tables give all three levels in numbers.
- **Section 2** — the full result of the document-intrusion validation, including the response-extraction accounting behind each accuracy figure.

Every number here is generated from the deposited data by `code/05_impact_analysis/supplementary_tables.py`; the same tables are also deposited as CSV (`supplementary/citation_shares_h1.csv` and its h2 and h3 counterparts, and all three levels together in `citation_shares_all_levels.csv`) for reuse.

1 Fractional citation shares by cluster

Impact is the *fractional citation sum*: a work's incoming citations are split among the countries that produced it, in proportion to the institutional shares recorded in `country_of_origin`, then summed within a cluster and grouped into blocs. China is `CN + HK + MO` (Taiwan is counted in RoW); the EU-27 is the 27 member states; RoW is every remaining country.

Shares are read **across a row**: the four bloc columns of a cluster sum to 100 %. The *% corpus* column instead reads down — it is that cluster's share of all fractional citations in the corpus, and the column sums to 100 % within each table. Meso- and macro-level figures are aggregations of the micro-level citation sums, not averages of micro-level shares, so a cluster's share is always its own citations over its own total.

Table 1: Fractional citation shares by bloc across the 5 macro-clusters (domains)

Code	Cluster	Frac. citations	% corpus	China	US	EU-27	RoW
0	Computer Science	4,454,326	22.48	31.54	19.54	15.50	33.43
1	Health Science	5,332,638	26.91	17.52	23.49	23.00	36.00
2	Social Science	2,985,947	15.07	12.97	15.72	25.60	45.71
3	Natural Science	3,564,219	17.99	23.05	16.86	23.58	36.51
4	Engineering	3,480,465	17.56	29.29	13.69	16.97	40.06

Table 2: Fractional citation shares by bloc across the 31 meso-clusters (fields)

Code	Cluster	Frac. citations	% corpus	China	US	EU-27	RoW
00	Computer Vision & Image Processing	1,934,964	9.76	42.23	16.34	12.75	28.68
01	Natural Language Processing	471,497	2.38	22.37	30.22	17.95	29.46
02	Speech Recognition & Audio Processing	282,246	1.42	20.51	20.47	18.60	40.42
03	Biomarkers for Health & Performance	564,789	2.85	22.72	12.98	19.58	44.72
04	Machine Learning Foundations	1,200,830	6.06	24.64	23.36	16.32	35.68

Code	Cluster	Frac. citations	% corpus	China	US	EU-27	RoW
10	Medical Imaging & Radiology	1,056,955	5.33	20.60	17.47	18.89	43.04
11	Neuroscience & Mental Health	888,886	4.49	11.34	25.38	28.90	34.38
12	Clinical Informatics & Public Health	1,337,947	6.75	11.36	23.73	22.28	42.63
13	Molecular Biology & Drug Discovery	569,563	2.87	16.99	26.69	21.60	34.72
14	Microbial and Immune Systems Biology	420,350	2.12	24.43	22.93	22.23	30.41
15	Genetics & Genomics	1,058,937	5.34	24.94	26.11	24.10	24.85
20	Education & Learning Science	712,487	3.60	11.53	15.36	20.40	52.71
21	Digital Media & Communication	448,176	2.26	15.39	21.03	16.33	47.25
22	Politics & Governance	175,571	0.89	5.31	22.02	31.52	41.15
23	Finance & Stock Market	200,951	1.01	15.84	15.00	22.16	47.00
24	Business & Management	1,003,006	5.06	14.91	12.16	29.31	43.62
25	Urban Development	251,347	1.27	13.34	10.52	34.55	41.59
26	Ethical & Creative AI	194,409	0.98	6.21	24.99	33.45	35.35
30	Climate & Hydrology	990,287	5.00	25.08	16.77	23.70	34.45
31	Material Science	960,067	4.84	25.74	17.00	22.26	34.99
32	Energy & Thermodynamics	189,063	0.95	19.98	9.11	18.78	52.13
33	Agriculture & Forestry	689,444	3.48	24.15	10.90	23.13	41.82
34	Astrophysics & Quantumphysics	322,702	1.63	7.07	31.82	33.59	27.53
35	Neuromorphic Hardware Accelerators	219,573	1.11	30.51	22.76	15.05	31.67
36	Construction & City Planning	193,083	0.97	16.55	13.74	28.74	40.97
40	Cybersecurity & Privacy	987,474	4.98	19.68	19.11	15.79	45.42
41	IoT & Edge Computing	672,163	3.39	33.41	10.85	16.88	38.86
42	Anomaly & Fault Detection	227,925	1.15	48.07	7.55	12.75	31.63
43	Energy Technologies	720,373	3.64	25.31	9.22	19.52	45.95
44	Autonomous Mobility	393,695	1.99	34.16	13.37	18.10	34.37
45	Robotics & Mechatronics	478,835	2.42	36.39	16.39	16.75	30.48

Table 3: Fractional citation shares by bloc across the 106 micro-clusters (research fronts)

Code	Cluster	Frac. citations	% corpus	China	US	EU-27	RoW
000	Image Generation & Segmentation	586,362	2.96	41.93	19.72	12.54	25.81
001	Face & Fingerprint Recognition	141,909	0.72	29.16	11.30	12.65	46.89
002	Multimodal Vision & Transformer	434,373	2.19	45.19	21.46	8.46	24.89
003	Object Detection & Classification	430,086	2.17	45.90	9.81	12.40	31.89
004	Mapping & Geospatial Analysis	342,234	1.73	39.82	14.33	19.03	26.82
010	Text to Text Transformer	246,892	1.25	18.68	32.06	17.52	31.73
011	Knowledge Graphs & Reasoning	224,605	1.13	26.42	28.19	18.42	26.97
020	Music & Acoustic Modeling	35,270	0.18	18.72	21.24	22.95	37.09
021	Emotion & Cognition in Audio	106,110	0.54	27.13	12.03	14.83	46.01
022	Speech Recognition	140,866	0.71	15.96	26.65	20.34	37.05
030	Signal Processing & Sensing	201,395	1.02	23.50	13.12	17.80	45.58
031	Human - Computer Interaction	363,394	1.83	22.28	12.91	20.57	44.24
040	Supervised Learning	277,724	1.40	36.30	21.86	11.56	30.29
041	Neural Network Architectures	218,340	1.10	22.97	29.88	14.84	32.31
042	SVM & Decision Tree	218,949	1.10	18.09	8.04	14.19	59.68
043	Probabilistic Modeling	225,010	1.14	15.54	31.96	23.26	29.25
044	Unsupervised Learning	144,624	0.73	12.48	26.06	26.52	34.93
045	Graph Learning & Network Modeling	116,183	0.59	45.05	23.53	8.37	23.05
100	Musculoskeletal Imaging	198,995	1.00	15.14	17.86	26.28	40.72
101	Neuroimaging & Imaging Physics	184,623	0.93	18.13	24.86	26.26	30.75
102	Cancer Detection & Screening	261,659	1.32	23.36	15.05	17.03	44.56
103	Disease Diagnostics	411,678	2.08	22.60	15.50	13.19	48.71
110	Neuroscience	501,730	2.53	12.64	27.40	27.78	32.18
111	Mental Health & Behaviorism	387,156	1.95	9.65	22.75	30.36	37.24
120	Cardiovascular Risk Modeling	167,296	0.84	16.47	23.83	28.34	31.36
121	Treatment Response & Risk Prediction	346,117	1.75	15.41	25.62	26.61	32.36

Code	Cluster	Frac. citations	% corpus	China	US	EU-27	RoW
122	Epidemiological Forecasting & Policy	173,747	0.88	13.25	26.08	17.77	42.91
123	Human-Centered Medical AI Systems	472,362	2.38	4.95	26.20	22.35	46.49
124	Clinical Decision Support	178,425	0.90	13.83	11.13	12.41	62.62
130	Molecular Biology	265,507	1.34	12.85	27.62	21.45	38.08
131	Drug Discovery	304,056	1.53	20.61	25.89	21.73	31.78
140	Infectious Disease & Immunology	272,256	1.37	23.41	22.51	20.53	33.55
141	Microbial Ecosystems	148,094	0.75	26.30	23.69	25.38	24.64
150	Cancer Genomics	409,147	2.06	33.63	21.07	23.42	21.88
151	Gene Regulation	264,434	1.33	17.07	36.10	22.44	24.39
152	Mutation & Evolutionary Genetics	126,771	0.64	11.46	34.20	21.96	32.38
153	Organ Physiology	258,585	1.30	25.83	19.92	27.93	26.33
200	Learning & Performance Analytics	200,153	1.01	17.91	11.16	18.21	52.72
201	Teaching & Curriculum Design	191,278	0.97	7.73	21.39	17.62	53.26
202	Learning Psychology & Motivation	187,424	0.95	9.87	15.71	24.05	50.38
203	Digital Learning Enhancement	133,632	0.67	9.74	12.53	22.54	55.19
210	Content Integrity & Safety	101,567	0.51	12.81	18.58	18.17	50.44
211	Social Media & Public Discourse	99,729	0.50	8.29	30.24	21.22	40.25
212	Sentiment Mining & Analysis	141,137	0.71	25.18	11.47	10.99	52.36
213	Dialogue & Conversational AI	105,743	0.53	11.49	27.46	17.06	44.00
220	International Politics & Law	89,653	0.45	3.34	21.26	32.01	43.39
221	Science Policy & Knowledge Networks	50,317	0.25	8.08	16.93	31.09	43.89
222	Game Theory & Decision Making	35,601	0.18	6.36	31.16	30.88	31.61
230	Market Forecasting & Algorithmic Trading	95,995	0.48	17.55	15.09	19.44	47.93
231	Digital Financial Services	65,526	0.33	14.09	11.79	23.94	50.18
232	Macroeconomic Financial Policy	39,430	0.20	14.58	20.12	25.83	39.48
240	Risk Analytics & Decision Modeling	146,888	0.74	11.50	17.53	29.36	41.61
241	Smart Manufacturing	207,935	1.05	12.03	10.18	37.83	39.95
242	Organizational Performance	130,777	0.66	10.32	11.27	30.25	48.16
243	Enterprise Digitization	242,765	1.22	8.78	9.39	34.11	47.73
244	Digital Marketing & Consumer Behavior	138,968	0.70	10.84	11.63	24.89	52.64
245	Psychographic Profiling & Personalization	135,673	0.68	42.57	15.74	11.25	30.44
250	Sustainable Development	222,005	1.12	12.51	11.44	34.80	41.26
251	Tourism	29,342	0.15	19.66	3.55	32.68	44.11
260	Artificial Cognition & Ethics	87,358	0.44	3.80	28.43	32.75	35.02
261	Digital Art	33,762	0.17	8.98	21.56	31.83	37.63
262	Virtual Reality	73,289	0.37	7.81	22.46	35.04	34.69
300	Tectonics & Geology	67,045	0.34	23.75	17.47	25.25	33.53
301	Topography & Earth-Observation	220,321	1.11	41.60	10.39	20.02	28.00
302	Hydrology & Marine Science	288,178	1.45	21.75	14.46	21.92	41.87
303	Ecosystem Ecology	115,831	0.58	6.82	24.05	30.37	38.75
304	Climate & Atmospheric Prediction	298,912	1.51	23.48	20.72	25.21	30.60
310	Flow, Wind & Heat Modeling	125,984	0.64	18.96	21.53	24.31	35.19
311	Material Strength, Damage & Fabrication	301,619	1.52	21.61	13.27	23.43	41.69
312	Polymers, Photonics & Biomaterials	532,464	2.69	29.69	18.05	21.11	31.15
320	Petroleum Mapping & Extraction	43,790	0.22	32.79	14.30	10.20	42.71
321	Environmental Engineering	145,273	0.73	16.12	7.55	21.36	54.97
330	Forest Ecosystems	183,986	0.93	26.00	13.31	25.68	35.01
331	Soil & Microbiome Monitoring	127,745	0.64	29.52	13.58	26.84	30.06
332	Crop Disease & Species Classifier	141,779	0.72	27.63	4.86	12.97	54.54
333	Livestock & Veterinary Science	81,423	0.41	17.37	11.06	29.33	42.23
334	Digital Agriculture & Yield Optimization	154,511	0.78	17.88	11.27	23.10	47.75
340	Astrophysics	123,157	0.62	5.48	30.93	35.88	27.70
341	Computational Quantum Systems	143,383	0.72	8.57	34.26	28.90	28.27
342	Experimental Particle & Nuclear Physics	56,162	0.28	6.71	27.50	40.53	25.26
350	Neuromorphic Computing	126,701	0.64	38.94	15.53	12.72	32.80
351	Circuit Architecture for Neural Systems	92,872	0.47	19.01	32.63	18.24	30.13
360	Construction Technology	64,213	0.32	15.62	8.41	32.05	43.92
361	Mobility & City Planning	128,870	0.65	17.01	16.40	27.09	39.50
400	Cybersecurity	462,739	2.33	19.53	15.44	14.77	50.27

Code	Cluster	Frac. citations	% corpus	China	US	EU-27	RoW
401	Blockchain & Cryptography	273,905	1.38	24.58	13.47	15.05	46.90
402	Hardware - Software Interaction	250,830	1.27	14.60	32.04	18.48	34.88
410	Wireless Communications	220,971	1.12	36.65	10.90	16.98	35.46
411	Edge Computing	312,430	1.58	38.79	10.42	15.33	35.46
412	Cloud & Distributed Systems	55,412	0.28	18.08	17.61	21.44	42.87
413	IoT Devices & Sensors	83,350	0.42	14.81	7.80	19.41	57.98
420	Structural Health Monitoring	53,475	0.27	42.18	8.13	16.34	33.35
421	Anomaly Detection	51,311	0.26	33.90	12.34	16.43	37.33
422	Fault Detection	123,139	0.62	56.54	5.30	9.66	28.51
430	NEV & Battery Health	74,794	0.38	30.48	7.28	24.30	37.95
431	Power Asset Monitoring	26,655	0.13	35.60	6.85	12.53	45.03
432	Smart Grid Analytics & Optimization	262,485	1.32	18.18	9.06	21.51	51.26
433	Energy System Modeling	233,931	1.18	27.82	11.12	17.66	43.40
434	Renewables & Grid Optimization	122,508	0.62	30.39	7.67	17.40	44.54
440	Autonomous Driving	153,439	0.77	27.27	12.21	21.05	39.48
441	Traffic Modeling	138,809	0.70	39.79	15.59	16.00	28.62
442	Networked Routing	101,447	0.51	36.88	12.08	16.52	34.52
450	Autonomous Navigation & Path Planning	101,346	0.51	43.36	11.10	15.24	30.30
451	Embodied Robotics	91,008	0.46	17.51	26.56	26.07	29.86
452	Task Scheduling & Decision Making	108,026	0.55	20.85	29.81	19.09	30.24
453	Control Systems & Fuzzy Logic	178,455	0.90	51.46	6.08	11.43	31.04

2 Document-intrusion validation

Each trial shows the judge a panel of five documents — four drawn from a target cluster, one intruder drawn from a different cluster at the same level, in randomised order — with titles and abstracts truncated to 200 words, and nothing else: no cluster label, no TF-IDF terms, no indication of which document is which. The judge names the intruder. The random-guess baseline is 20 %.

Table 4 gives the three reported runs. *Recovered* counts responses whose verdict was read out of the reasoning trace because the model never wrote the requested final line; *forced choice* counts responses that hit the token ceiling mid-reasoning and were resolved by presenting the truncated trace to a second, non-reasoning model for extraction only. Neither step guesses: no trial was scored by fallback to a random answer, and no response went unparsed.

Level	Granularity	Trials	Correct	Accuracy	Recovered	Forced choice
h1	5 domains	1,000	405	40.5 %	22	103
h2	31 fields	1,000	647	64.7 %	11	70
h3	106 research fronts	1,000	778	77.8 %	7	37

Table 4: Document-intrusion detection, 1,000 trials per hierarchy level. Model `deepseek-v4-flash` with reasoning enabled, `decisive` prompt variant, 200-word truncation, five-document panels, seed 20250628. Random-guess baseline 20 %.

Accuracy rises monotonically as the taxonomy becomes finer, which is what the protocol predicts: intrusion detection depends on the ratio of within-cluster spread to between-cluster separation, and a macro-domain is heterogeneous by construction. A panel drawn from `Natural Science` may pair structural health monitoring with ionospheric prediction and rural land-use change; an intruder is far less conspicuous there than among the narrow, lexically distinctive research fronts at h3.

The complete record of all 3,000 trials — the panel shown, the model's full reasoning, the extracted verdict and the rule that extracted it — is deposited under `results/intrusion/`, one manifest (`.json`), one record per trial (`.jsonl`) and one readable transcript (`.txt`) per level.